

Report
Of
Full Stack Developer Intern

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: CS-2341805

Name of Student: Divyanshu Tiwari

Batch: 2023-2027

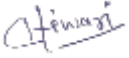
2026-27

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CANDIDATE'S DECLARATION

I, Divyanshu Tiwari do hereby declare that the internship report titled “Full Stack Developer Intern” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: 

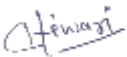
Student Name: Divyanshu Tiwari

Roll No: CS-2341805

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature: 

Student Name: Divyanshu Tiwari

Roll No: CS-2341805

INTERNSHIP COMPLETION CERTIFICATE

The certificate below, issued by Code-B Solutions Pvt. Ltd. (CIN: U62013MH2023PTC399457), certifies that Divyanshu Tiwari successfully completed his internship as a Full Stack Developer Intern, contributing from 09-03-2026 to 01-08-2026. The certificate is signed by Code-B Solutions Pvt. Ltd. and was generated on 01-08-2026.



CIN : U62013MH2023PTC399457

Code B Solutions Pvt.Ltd, Bungalow No-23, RDP 2, MHADA Colony, Mulund East, Mumbai 400081

Dear Divyanshu Tiwari,

We are pleased to formally acknowledge the successful completion of your internship as an **Full Stack Developer Intern** at CODE-B SOLUTIONS PRIVATE LIMITED. Your dedication and contributions from 09-03-2026 to 01-08-2026 have been greatly appreciated.

During your time with us, you demonstrated a strong understanding of Java and its Framework Spring Boot and also a commitment towards learning and collaboration. Your efforts have contributed positively to our projects, and we commend your growth throughout this experience. Sincerely,

Code-B Solutions Pvt Ltd.

CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

This report presents the work carried out during a six-month internship undertaken at Code-B Solutions Pvt. Ltd., Mumbai (Remote), from March 2026 to August 2026, in the role of Java Backend & Cloud Engineering Intern (designated as Full Stack Developer Intern on the internship completion certificate). The internship focused on backend development, REST API design, cloud deployment, and DevOps practices, bridging academic concepts with production-oriented software engineering by designing, developing, testing, and deploying scalable backend services using Java and Spring Boot.

4.2 Organization Profile

Code-B Solutions Pvt. Ltd. (CIN: U62013MH2023PTC399457) is a Mumbai-based technology company headquartered at Bungalow No-23, RDP 2, MHADA Colony, Mulund East, Mumbai 400081. The organization builds backend and cloud-based software solutions and offers structured internship programs that expose students to production-grade engineering practices, including API design, database optimization, containerization, and continuous integration and deployment workflows.

4.3 Problem Statement

Undergraduate students typically learn Java, object-oriented programming, and database concepts in isolated academic courses, with limited exposure to how these fundamentals translate into production-grade backend systems that must be reliable, scalable, and deployable on the cloud. This internship was undertaken to bridge that gap by providing hands-on experience in building REST APIs, optimizing databases for performance, and deploying and automating backend services using modern DevOps tooling.

4.4 Project Objectives

- Strengthen Java & OOP Design — deepen programming fundamentals and object-oriented principles.
- Production-Grade REST APIs — build APIs using Spring Boot with real-world reliability standards.

- Scalable Backend Architecture — understand distributed systems and scalable service design.
- Cloud & DevOps Mastery — deploy on AWS EC2, containerize with Docker, and automate CI/CD.

4.5 Scope of the Project

The scope of the internship covered the complete backend engineering lifecycle: designing and implementing REST APIs using Java and Spring Boot; securing services with Spring Security; persisting and querying data through Spring Data JPA and PostgreSQL; and deploying the resulting services to AWS EC2 using Docker containers and an automated CI/CD pipeline built with GitHub Actions. The scope also included quality engineering practices such as unit testing, integration testing, and API contract testing to ensure reliability across distributed service boundaries.

4.6 Technologies and Tools Used

- Backend & Framework: Java, Spring Boot, Spring Security, Spring Data JPA, RESTful APIs
- Database & Cloud: PostgreSQL, AWS EC2
- DevOps & Containers: Docker, GitHub Actions, CI/CD, Git, Maven, Linux
- Testing & API Tools: JUnit, Integration Testing, API Contract Testing, Postman
- Project Management: JIRA

4.7 System Architecture

The backend services followed a layered architecture in which an incoming client request is first received by the REST API layer, implemented through Spring Boot controllers that perform request validation, and is then passed to the service layer, which handles business logic and orchestration before interacting with the PostgreSQL database. For deployment, the system used an automated pipeline: code pushed to the Git repository automatically triggers a GitHub Actions CI/CD pipeline, which containerizes the services using Docker for consistent environments and deploys the production backend to AWS EC2. This automated workflow improved deployment efficiency and consistency while significantly reducing manual intervention in production releases.

4.8 Methodology

Development followed an iterative, production-oriented methodology. More than 15 production-grade REST APIs were designed and deployed using Spring Boot, with centralized exception handling and input validation implemented consistently across all endpoints, and consistent API response contracts maintained for reliable client integration, applying Domain-Driven Design and clean engineering principles across distributed service boundaries. A significant focus area was database optimization: high query latency that was degrading backend response times and user experience was addressed through schema optimization, advanced indexing strategies, and query execution plan analysis based on PostgreSQL best practices for distributed queries, resulting in an approximately 45% reduction in database query latency. Quality was embedded throughout the development lifecycle through centralized error handling, comprehensive input validation at all API boundaries, JUnit-based unit and integration testing, and API contract testing to prevent breaking changes across services, together delivering an approximately 30% improvement in backend reliability. Cloud deployment and DevOps practices were applied by containerizing services with Docker and automating build, test, and deployment through a GitHub Actions CI/CD pipeline targeting AWS EC2.

4.9 Expected Outcomes

- 15+ production-grade REST API endpoints designed and deployed.
- 10K+ monthly requests served reliably across distributed services.
- ~45% reduction in database query latency through optimization and indexing.
- ~30% improvement in backend reliability through quality engineering practices.

Beyond these measurable outcomes, the internship strengthened production-grade backend architecture and scalable system design skills, advanced database optimization and performance-tuning capability, and hands-on proficiency in cloud deployment, containerization, and DevOps workflows. It also built professional practices in API design, testing, and reliability engineering, providing a strong foundation for a career in backend, cloud, and distributed systems engineering.

4.10 Certificates of Completion and Communication Proof

The Internship Completion Certificate issued by Code-B Solutions Pvt. Ltd. to Divyanshu Tiwari, confirming contributions from 09-03-2026 to 01-08-2026 as a Full Stack Developer Intern and generated on 01-08-2026, is presented in the Internship Completion Certificate section of this report (front matter).

CHAPTER 5: BIBLIOGRAPHY/REFERENCES

1. Code-B Solutions Pvt. Ltd. — Summer Internship Program, Java Backend & Cloud Engineering, 2026.
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